

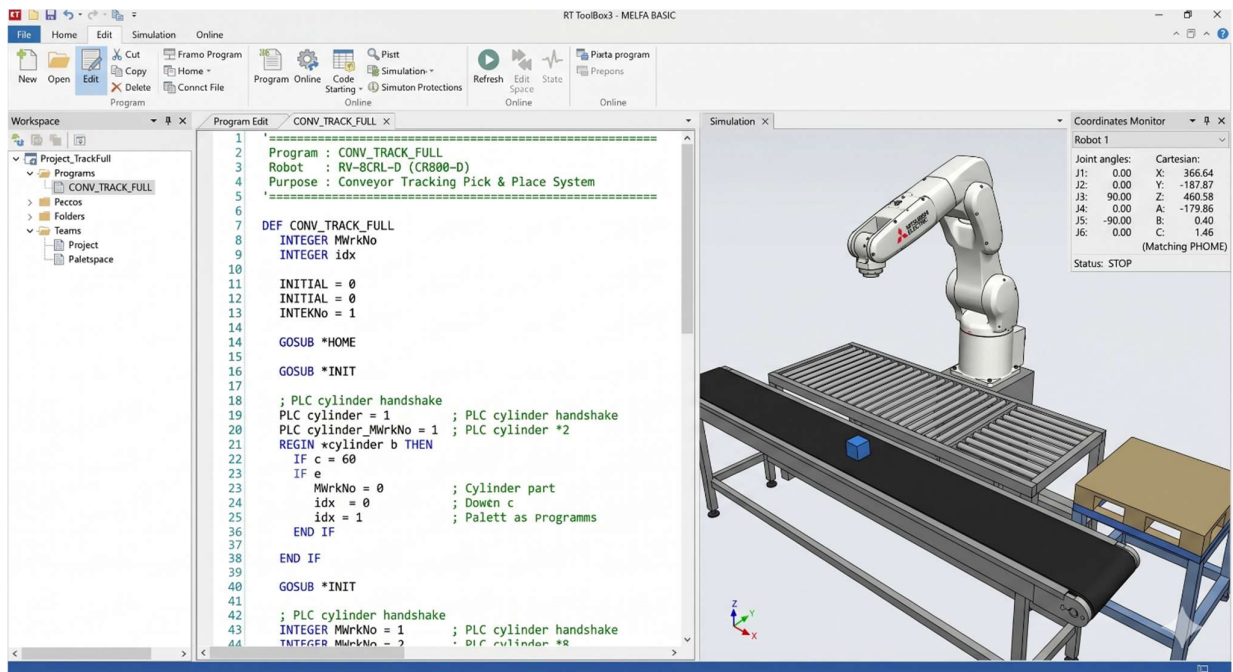


Figure 1: Program Initialization and Start of the 'CONV_TRACK_FULL' Loop.

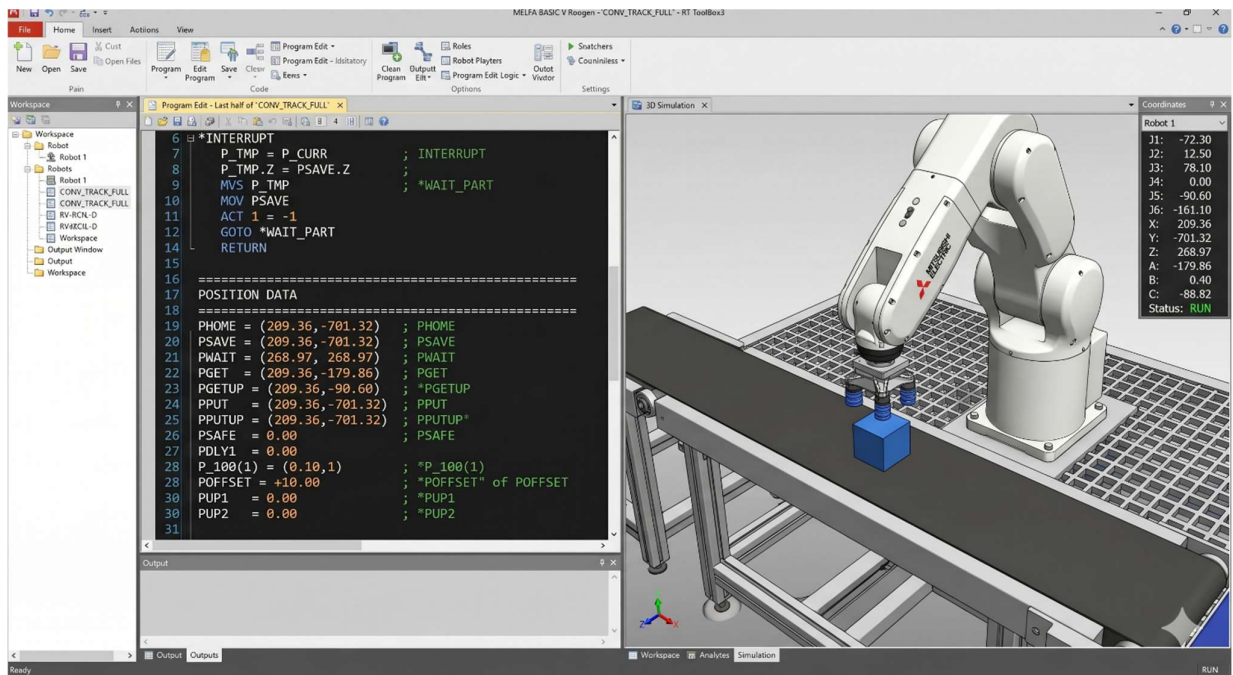


Figure 2: Complete Defined Position Coordinates and Variables Data Block.

Expt No:

06

Date:

12 / 03 / 2026

**AUTONOMOUS MATERIAL HANDLING
AND MANIPULATION LOGIC**

1. AIM:

To design and implement an automated pick-and-place robotic sequence using asynchronous sensor handshaking, effector actuation, and precise spatial trajectory mapping on a 6-axis industrial manipulator.

2. APPARATUS REQUIRED:

- a) Robotic System: Mitsubishi RV-8CRL-D industrial manipulator.
- b) Control Hardware: CR800-Series Controller.
- c) Software Suite: RT ToolBox3 integrated engineering environment.
- d) Sensing Hardware: Photoelectric workpiece detection sensor mapped to Input 6031.
- e) Actuators: Component transport conveyor and pneumatic suction/gripper assembly actuated via Outputs 12/13.

3. THEORY:

A fundamental robotic pick-and-place operation requires the seamless integration of spatial motion planning with digital logic control to interact with the external environment.

a) Core Principle:

Autonomous manipulation relies on event-driven execution. The robot remains in a standby state until an external physical sensor triggers the initiation of a predefined sequence of kinematic sub-routines (retrieval, transfer, and placement).

b) Working Mechanism:

- i) Approach/Escape Kinematics: To prevent lateral shear collisions with fixtures or components during retrieval and placement, the kinematic sequence strictly enforces relative Z-axis offsets (approach and escape heights). This ensures the tool only enters the manipulation zone vertically.
- ii) I/O Synchronization: The program architecture utilizes Wait M_In logic to halt motion execution until workpiece presence is verified by external instrumentation. Subsequent M_Out signals are triggered to electronically actuate pneumatic valves for payload securing and release.
- iii) Point-to-Point Transfer: The sequence utilizes a combination of joint interpolation (Mov) for rapid spatial repositioning and linear interpolation (Mvs) for the precise vertical insertion into the pick and place coordinates.

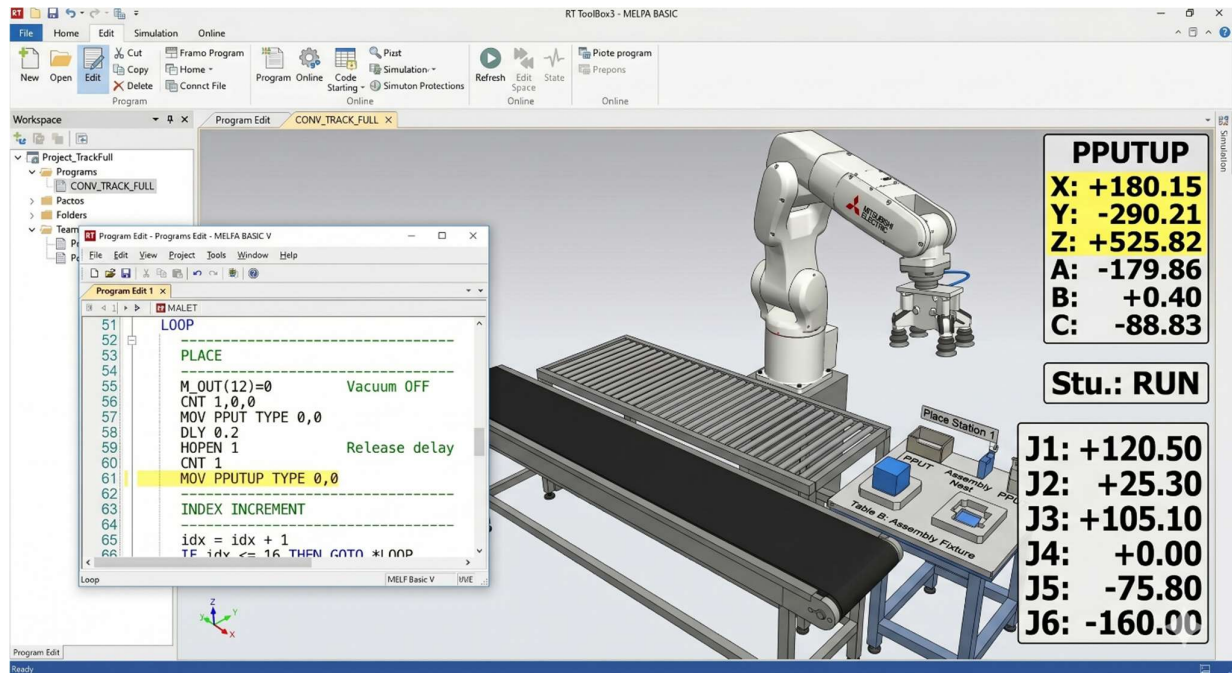


Figure 3: Maximized 3D Simulation Pane with Real-Time Coordinate Overlay at 'PGET' (Conveyor) Pick Station.

4. PROCEDURE:

a) Setup & Connections:

Interface the photoelectric sensor and pneumatic solenoid valves with the controller's I/O terminals. Configure the conveyor system and define the absolute spatial coordinates for the "Pick" (pPick) and "Place" (pPlace) stations.

b) Algorithm & Logic:

- i) Activate the component conveyor and enter a continuous monitor loop for workpiece arrival.
- ii) Upon photoelectric detection, halt the conveyor to stabilize the workpiece.
- iii) Execute the kinematic "Pick" sequence: move to a safe Z-offset above the part, perform a linear approach, engage the pneumatic effector, and execute a vertical escape.
- iv) Execute the "Place" sequence: move to a safe Z-offset above the drop zone, perform a linear insertion, disengage the pneumatic effector, and execute a vertical escape.
- v) Reactivate the conveyor and loop the process for continuous automation.

c) Execution Steps:

- i) Teach and record the absolute coordinates for pPick and pPlace using the teaching pendant.
- ii) Verify the "Hand Type" parameter in RT ToolBox3 matches the installed pneumatic assembly logic.
- iii) Validate the digital I/O handshake mechanism utilizing the Signal Monitor screen.
- iv) Execute a Step Run to ensure the Z-offsets safely clear all environmental obstacles before switching to Automatic mode for full-speed execution.

d) Observation Instructions:

Analyze the operational Tact Time (cycle time from pick to place) and monitor the pneumatic grip stability during rapid joint interpolation (Mov) between the pick and place stations.

5. RESULT AND OBSERVATION:

The robotic system successfully performed synchronized pick-and-place operations with external conveyor and sensor systems. Proper approach/escape height control prevented interference, and reliable sensor handshaking ensured accurate part handling, as verified by faculty.